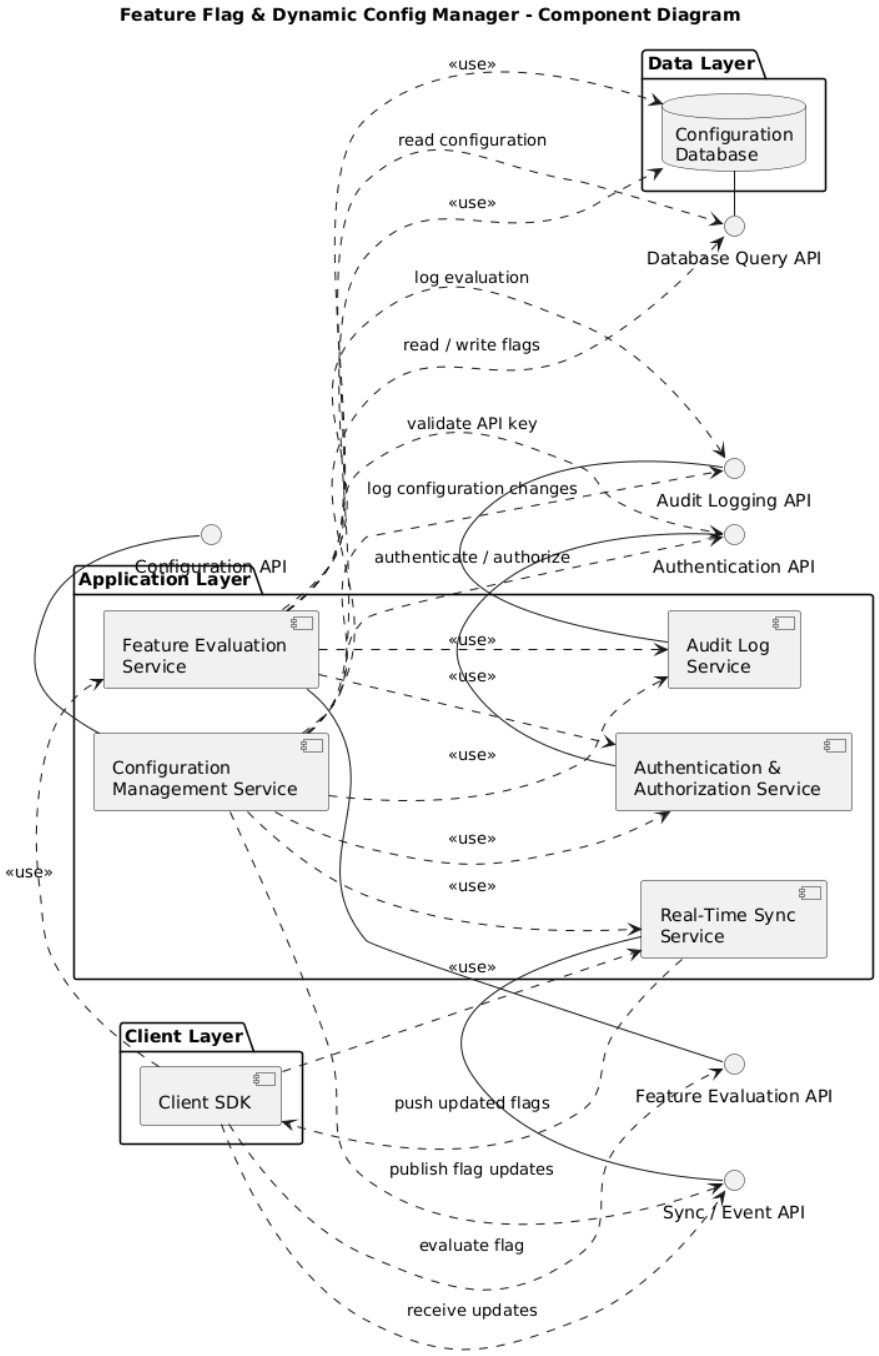
Software Engineering (UE24CS341A)

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Architecture Justification – Feature Flag & Dynamic Config Manager



Selected Architecture: Layered Architecture

The Feature Flag & Dynamic Config Manager uses a layered architecture consisting of Client, Business, and Data layers. The Client layer contains the Client SDK. The Business layer contains the Feature Evaluation Service, Configuration Management Service, Authentication & Authorization Service, Real-Time Sync Service, and Audit Log Service. The Data layer contains the Configuration Database.

Scenario-related reasons

1. **Separation of responsibilities:** Feature evaluation, configuration management, authentication, real-time synchronization, and auditing are separated into individual components. This makes the system easier to understand, maintain, and modify as new feature-flag requirements are introduced.
2. **Clear data and service flow:** The layered structure separates client requests from business logic and data storage. The Client SDK communicates with business services, while configuration data is handled through the Configuration Database. This supports environment-specific configuration and reduces the chance of configuration logic being mixed with data-access logic.

Security advantage

Authentication and Authorization is isolated as a dedicated service in the Business layer. This allows production modifications to be restricted to authenticated Release Managers and prevents unauthorized configuration changes. It also provides a clear security boundary for access to configuration-management operations.

Performance benefit

Separating the Feature Evaluation Service from configuration-management operations allows flag evaluation to follow a direct path from the Client SDK to the evaluation logic and configuration data. This supports the requirement that the evaluation API respond within 15 ms. Real-Time Sync can also push updated flag states to connected SDKs without requiring client application restarts.